

Unit A Practice Problems: Joint Distribution

Instructor: Dr. Md Mutasim Billah

Required Practice Problems

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. To prepare for the exams, make sure you have completed and reviewed all the required practice problems from the unit. The **solutions** are posted on Canvas (`textbook-solution.pdf` > Practice Problem-Solution > STAT 312 > Files).]

Solve the following exercises from the “Chapter 5 Exercises” section of the textbook (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger).

- 5.1.1, 5.1.2, 5.1.9, 5.2.2, 5.4.5 (find $\text{cov}(X,Y)$ only).

[To find these exercises in the digital copy of the textbook (Textbooks > STAT 312 > Files > Canvas), go to the xvi Contents page and click the “EXERCISES” tab located at the lower-right corner of the page. This will open the complete list of chapter exercises.

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.

Chapter 5 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 5.4.1 (find $\text{cov}(X,Y)$ only).